

# JEA Production Architecture

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# JEA Production System Architecture

**Document:** System Architecture Design for JEA Digital Services Platform

**Environment:** Production

**Prepared:** 2026-07-25

**Scale:** 90,000 members · 80,000 buildings/year · ~15 documents per building

## 1. Executive Summary

مُصمَّمة (JEA Digital Services Platform) هذه الوثيقة تُقدِّم معمارية إنتاج كاملة لنظام نقابة المهندسين الأردنيين لخدمة 90 ألف عضو و 80 ألف مبنى سنوياً (~220 معاملة/يوم) مع دعم رفع ~1.2 مليون مستند سنوياً.

نمو أفقي — **Horizontal Scale** — لا نقطة فشل واحدة (No SPOF) — **High Availability** — المبادئ الأساسية  
— **Regulatory Compliance** — blue/green + rolling — **Zero-Downtime Deployments** — للتطبيق والقراءة  
CSP, WAF, audit trail, MODEE Annex 4.15 (GSB) — **Disaster Recovery** — RPO ≤ 15min, RTO ≤ 1h

## 2. Scale Requirements

| المتطلب             | القيمة | الأثر التصميمي                             |
|---------------------|--------|--------------------------------------------|
| المستخدمون المسجلون | 90,000 | Auth cache, session store, user management |

| المتطلب                       | القيمة             | الأثر التصميمي                              |
|-------------------------------|--------------------|---------------------------------------------|
| مبانٍ سنوياً                  | 80,000             | ~220/day, ~9/hour peak                      |
| مستندات سنوياً                | ~1.2M              | Object storage 1-2 TB in year 1, grow 30%/y |
| متزامنون في الذروة            | ~500-1,000         | 3-5 app servers minimum + auto-scale        |
| Reviewer decisions/day        | ~200-400           | Read-heavy dashboard queries → replicas     |
| Peak concurrent uploads       | 20-50              | Streaming to object storage                 |
| Certificate verifications/day | 500-2000+ (public) | CDN critical                                |

### 3. Architecture Overview

#### 3.1 High-Level Diagram (Mermaid)

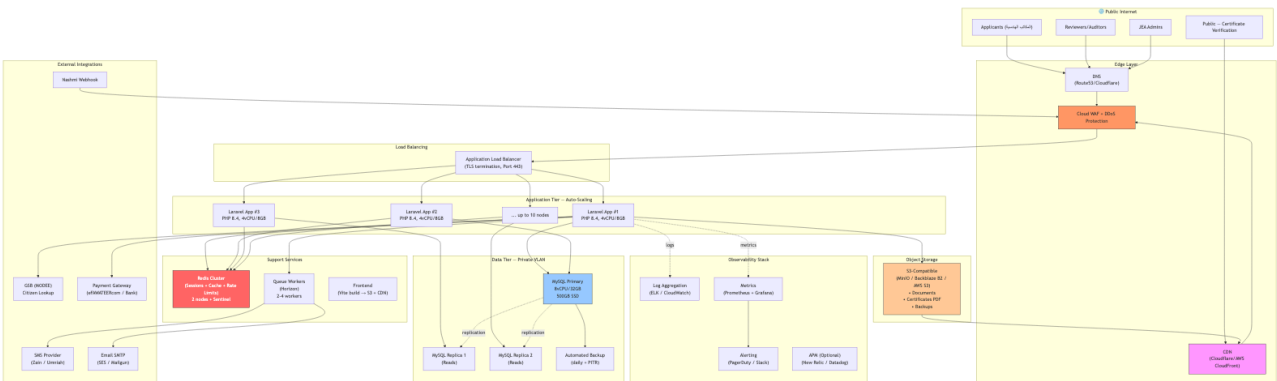
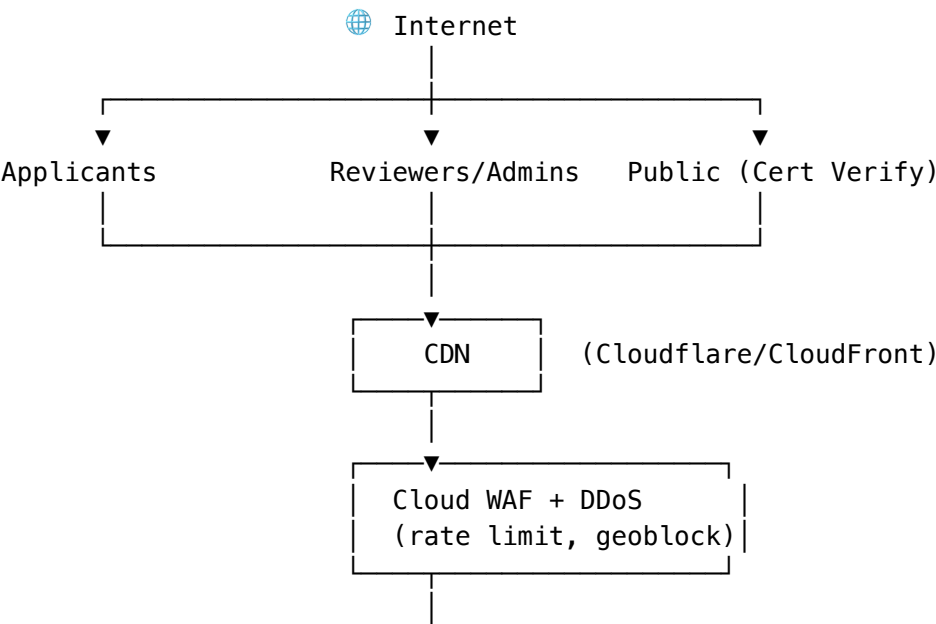
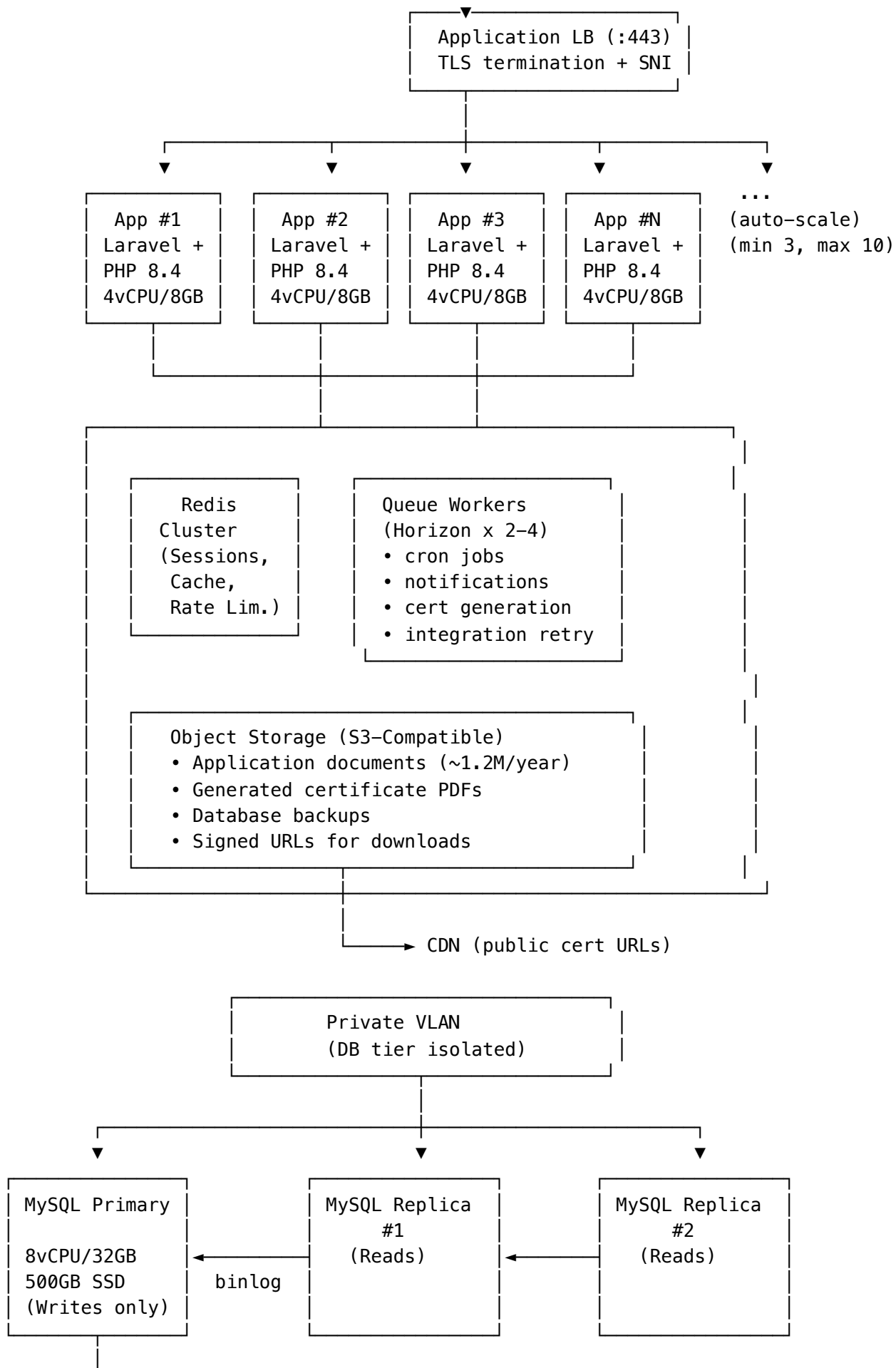


Diagram 1

#### 3.2 ASCII Overview (Fallback)







#### Automated Backup

- Full backup daily (S3 encrypted)
- Point-in-Time Recovery (binlog shipping)
- Retention: 30 days
- Cross-region copy (DR)

External Integrations (all through the App tier via HTTPS):

|                 |                                            |
|-----------------|--------------------------------------------|
| GSB (MODEE)     | – Citizen data lookup (OTP + IP whitelist) |
| Nashmi Webhook  | – Bidirectional (X-Integration-Key)        |
| Payment Gateway | – eFAWATEERcom / Arab Bank                 |
| SMS Provider    | – Zain / Umniah / Orange                   |
| Email SMTP      | – AWS SES / Mailgun                        |

Observability Stack:

|         |                                                                  |
|---------|------------------------------------------------------------------|
| Logs    | → ELK (Elasticsearch/Logstash/Kibana) or AWS CloudWatch Logs     |
| Metrics | → Prometheus + Grafana (P95 latency, error rate, throughput, DB) |
| Alerts  | → PagerDuty / Slack (on-call rotation)                           |
| APM     | → New Relic / Datadog (optional)                                 |
| Uptime  | → Statuspage.io / Better Uptime                                  |

## 4. Component Specifications

### 4.1 Edge Layer

| Component       | Technology               | Purpose                            | Sizing                 |
|-----------------|--------------------------|------------------------------------|------------------------|
| DNS             | Route53 / Cloudflare DNS | Domain resolution, health checks   | Standard               |
| CDN             | Cloudflare / CloudFront  | Static assets, public cert PDFs    | Pro tier (\$20-100/mo) |
| WAF             | Cloudflare WAF / AWS WAF | OWASP top 10, rate limit, geoblock | Managed rules          |
| DDoS Protection | Cloudflare / AWS Shield  | Volumetric + application-layer     | Standard/Advanced      |

### 4.2 Application Tier

| Component       | Technology                           | Sizing                        | Notes                                |
|-----------------|--------------------------------------|-------------------------------|--------------------------------------|
| Load Balancer   | AWS ALB / nginx                      | Health checks every 15s       | Sticky sessions OFF (stateless)      |
| App Server      | Laravel 13 + PHP 8.4 + Nginx/PHP-FPM | 4 vCPU / 8 GB RAM / 50 GB SSD | Min 3, max 10, auto-scale on CPU>70% |
| Frontend        | Vite build → static files on CDN     | N/A                           | Served from S3+CDN, not app tier     |
| Deploy Strategy | Blue/Green or Rolling                | Zero downtime                 | Health checks gate traffic shift     |

### 4.3 Data Tier

| Component           | Technology                    | Sizing                                | Notes                                |
|---------------------|-------------------------------|---------------------------------------|--------------------------------------|
| MySQL Primary       | MySQL 8.0 / MariaDB           | 8 vCPU / 32 GB RAM / 500 GB SSD (gp3) | Writes only, IOPS 3000+              |
| MySQL Read Replicas | MySQL 8.0 (async replication) | 4 vCPU / 16 GB RAM / 500 GB SSD × 2   | Reviewer dashboards, reports         |
| Backups             | mysqldump + binlog to S3      | Full daily + PITR                     | Encrypted at rest, cross-region copy |
| Redis Cluster       | Redis 7 + Sentinel            | 2 vCPU / 4 GB × 2 nodes               | Sessions, cache, rate limits, queue  |
| Queue Workers       | Laravel Horizon               | 2-4 worker nodes (2 vCPU / 4 GB each) | Notifications, cron, cert generation |

### 4.4 Storage

| Component      | Technology                    | Capacity                | Notes                              |
|----------------|-------------------------------|-------------------------|------------------------------------|
| Object Storage | MinIO / Backblaze B2 / AWS S3 | Start 1 TB, grow ~30%/y | Documents + certificates + backups |

| Component | Technology | Capacity     | Notes                                  |
|-----------|------------|--------------|----------------------------------------|
| CDN Cache | Cloudflare | Auto-managed | Serves signed URLs from object storage |

### 4.5 External Integrations

| Integration     | Purpose                                              | Auth                              | Handled by                 |
|-----------------|------------------------------------------------------|-----------------------------------|----------------------------|
| GSB (MODEE)     | Citizen data lookup (name, ID lookup for compliance) | OTP + IP whitelist (§4.5 rule 11) | Integrations\Gsb\          |
| Nashmi          | Contractor management webhook (bidirectional)        | X-Integration-Key header          | Integrations\Nashmi\       |
| Payment Gateway | Fee collection (eFAWATEERcom, banks)                 | API key + HMAC signature          | New: Integrations\Payment\ |
| SMS             | OTP + notifications (Arabic)                         | API key                           | Provider abstraction       |
| Email SMTP      | Transactional email (Arabic + English)               | SMTP creds                        | Laravel Mail               |

## 5. Data Flow Diagrams

### 5.1 Application Submission Flow

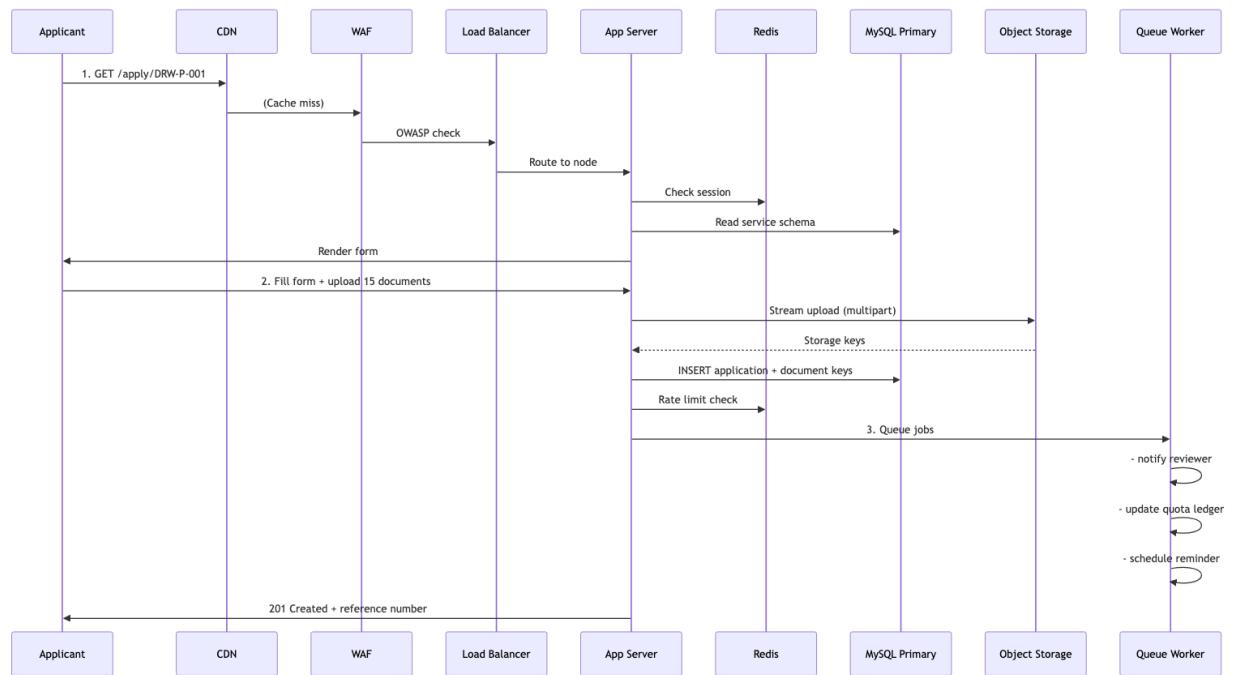


Diagram 2

## 5.2 Public Certificate Verification (High-Volume Path)

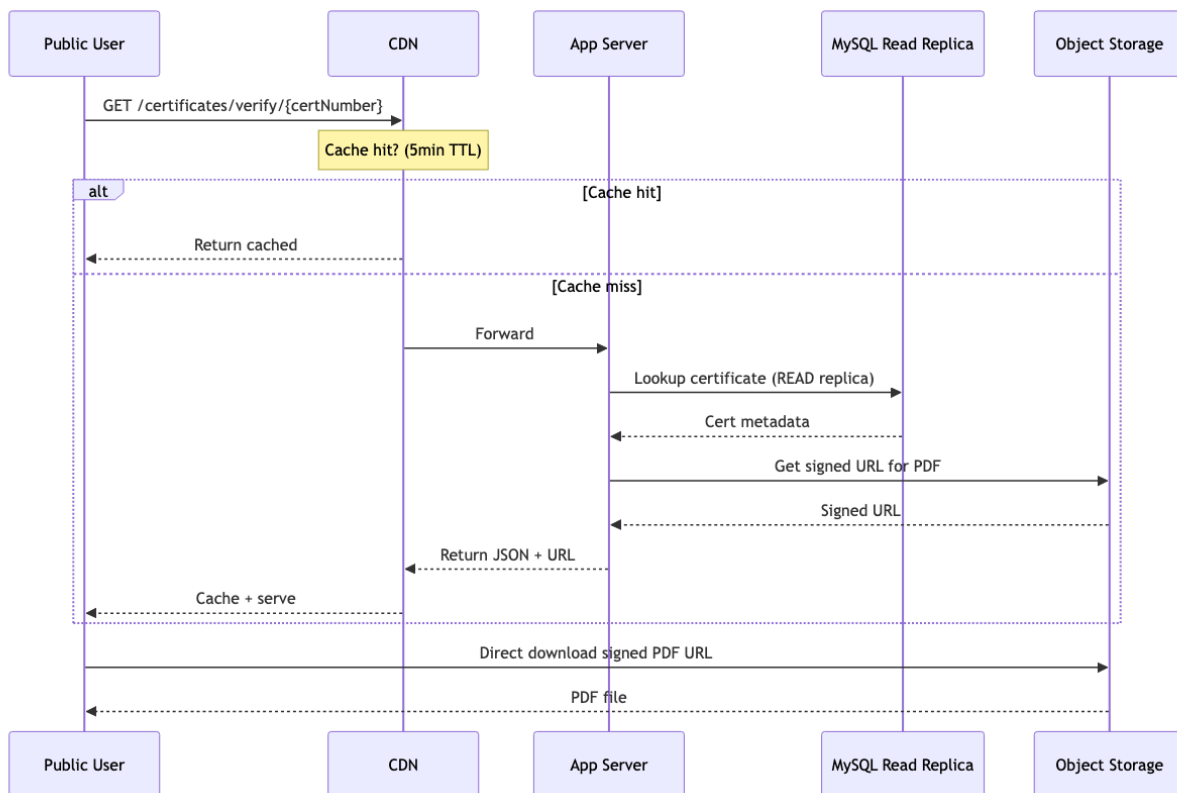


Diagram 3

## 5.3 Integration Flow — GSB Citizen Lookup

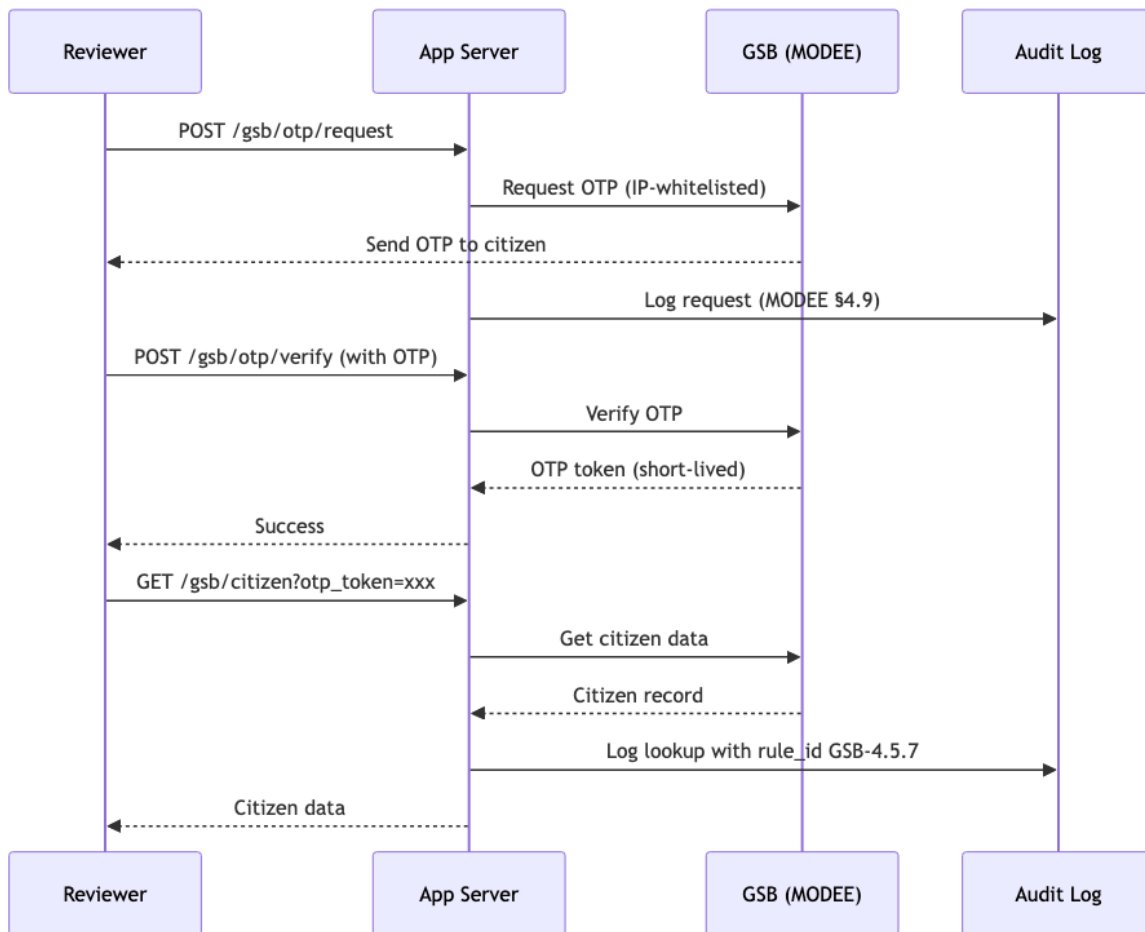


Diagram 4

## 6. Security Model

### 6.1 Defense in Depth

| Layer          | Control                | Implementation                                                 |
|----------------|------------------------|----------------------------------------------------------------|
| 1. Edge        | DDoS + WAF             | Cloudflare/AWS Shield + managed OWASP rules                    |
| 2. Transport   | TLS 1.3 only           | LB terminates, HSTS, cert auto-renewal                         |
| 3. Application | Authentication         | Sanctum + httpOnly cookies + CSRF                              |
| 3. Application | Authorization          | Role-based ( <code>CheckRole</code> middleware) + policy layer |
| 3. Application | Input validation       | <code>FormRequest</code> + <code>SchemaValidator</code>        |
| 3. Application | Rate limiting          | <code>throttle:*</code> limiters with Redis backend            |
| 3. Application | Session security       | 30-min inactivity timeout + rotation                           |
| 3. Application | CSP + Security Headers | <code>SecurityHeaders</code> middleware                        |



| Layer     | Control               | Implementation                          |
|-----------|-----------------------|-----------------------------------------|
| 4. Data   | Encryption at rest    | RDS/EBS encryption + S3 SSE             |
| 4. Data   | Encryption in transit | TLS to DB, private VLAN                 |
| 4. Data   | Backup encryption     | S3-managed keys + cross-region copy     |
| 5. Audit  | Full audit trail      | AuditLog table + rule_id per action     |
| 5. Audit  | Log aggregation       | ELK/CloudWatch retention 90 days min    |
| 6. Access | IAM + IP whitelist    | GSB IP whitelist enforced by middleware |
| 6. Access | Secrets management    | AWS Secrets Manager / HashiCorp Vault   |

### 6.2 Compliance Checklist

-  **MODEE Annex 4.15** — GSB IP whitelist + audit log
-  **JEA data protection** — Encrypted at rest + in transit
-  **OWASP Top 10** — WAF + input validation + parameterized queries
-  **Jordan Data Protection Law 2023** — Consent, retention limits, right-to-be-forgotten (add if needed)
-  **PCI-DSS** (if handling card data directly) — probably not, payment gateway handles it

## 7. High Availability & Disaster Recovery

### 7.1 HA Targets

| Metric                         | Target                                |
|--------------------------------|---------------------------------------|
| Uptime SLA                     | 99.9% (43 min downtime/month allowed) |
| RPO (Recovery Point Objective) | ≤ 15 minutes                          |
| RTO (Recovery Time Objective)  | ≤ 1 hour                              |

### 7.2 Redundancy Matrix

| Component   | Redundancy                     | Failure Scenario                          |
|-------------|--------------------------------|-------------------------------------------|
| App servers | Auto-scale group 3+ across AZs | Any node dies → LB removes, auto-replaces |

| Component      | Redundancy                                    | Failure Scenario                        |
|----------------|-----------------------------------------------|-----------------------------------------|
| DB Primary     | Automated snapshot every 6h + binlog shipping | Failover to replica in <5min            |
| DB Replicas    | 2× replicas                                   | Reviewer dashboards degrade gracefully  |
| Redis          | Master + Sentinel + 1 replica                 | Sentinel promotes replica automatically |
| Object Storage | S3 multi-AZ (11 nines durability)             | Data essentially indestructible         |
| CDN            | Multi-POP (global)                            | Regional failure → auto-reroute         |
| Nashmi/GSB     | Retry with exponential backoff                | Circuit breaker after N failures        |

## 7.3 DR Plan (Regional Failure)

1. **Detection** — health check fails 3× consecutive from 2+ regions
2. **Failover** — DNS switches to DR region (TTL 60s)
3. **Data** — DR region has async replica (RPO 15min max)
4. **App** — DR region has warm standby (3 min to full scale)
5. **Verification** — smoke tests + manual review before public

# 8. Monitoring & Observability

## 8.1 Golden Signals (Per Service)

- **Latency** — P50, P95, P99 (target: P95 < 500ms)
- **Traffic** — Requests/sec per endpoint
- **Errors** — 5xx rate < 0.1%, 4xx trend
- **Saturation** — CPU, memory, DB connections, queue depth

## 8.2 Dashboards (Grafana)

| Dashboard          | Metrics                                          |
|--------------------|--------------------------------------------------|
| Application Health | Request rate, error rate, latency, active users  |
| Database           | Connections, slow queries, replication lag, disk |

| Dashboard      | Metrics                                               |
|----------------|-------------------------------------------------------|
| Queue          | Jobs processed, failed jobs, queue depth by class     |
| Business       | Applications/day, decisions/day, cert issued, revenue |
| Integrations   | GSB call success rate, Nashmi webhook lag             |
| Infrastructure | CPU, memory, network, disk I/O per node               |

### 8.3 Alerting Thresholds

| Severity    | Condition                                 | Action                          |
|-------------|-------------------------------------------|---------------------------------|
| P1 Critical | 5xx rate > 5% for 2min OR DB primary down | Page on-call + incident channel |
| P2 High     | P95 latency > 2s for 5min OR queue > 1000 | Slack alert + investigate       |
| P3 Warning  | Disk > 80% OR replica lag > 30s           | Ticket for review               |
| P4 Info     | Deploy events, cron completions           | Log only                        |

## 9. Cost Estimate (Monthly, Rough)

Assuming AWS-equivalent pricing in JOD (USD × 0.71).

| Component                              | Est. USD/mo | Notes                        |
|----------------------------------------|-------------|------------------------------|
| App servers (3-5 × m5.large avg)       | \$250-400   | Auto-scale, spot for staging |
| DB Primary (RDS db.m5.2xlarge)         | \$500       | Reserved 1-year: -30%        |
| DB Replicas × 2 (db.m5.xlarge)         | \$500       | Reserved 1-year: -30%        |
| Redis (ElastiCache cache.m5.large × 2) | \$200       |                              |
| Object Storage (1-2 TB + egress)       | \$50-150    | Grows with usage             |

| Component                          | Est. USD/mo                                 | Notes                              |
|------------------------------------|---------------------------------------------|------------------------------------|
| CDN (Cloudflare Pro)               | \$20-100                                    | Or Cloudflare Free for low traffic |
| WAF                                | \$50-100                                    | Managed rules                      |
| Load Balancer                      | \$25                                        | ALB base + LCU                     |
| Backup storage (S3 + cross-region) | \$30-80                                     | 500 GB × 30 days retention         |
| Monitoring (CloudWatch/Datadog)    | \$100-300                                   | Depends on retention               |
| SMS + Email                        | \$200-800                                   | Volume-dependent                   |
| <b>Total (year 1)</b>              | <b>~\$1,900-2,700/mo (~JOD 1,400-1,900)</b> | Excluding one-time setup           |

*JOD equivalent for Hostinger/local hosting will differ. Above assumes public cloud.*

## 10. Deployment Phases

### Phase 0 — Prerequisites (Week 1)

- Provision AWS/cloud account + billing setup
- DNS + domain (jea.jo or similar)
- SSL certificates (Let's Encrypt or paid CA)

### Phase 1 — Foundation (Weeks 2-3)

- VPC, subnets (public + private + isolated)
- IAM roles + secrets management
- CI/CD pipeline (GitHub Actions → deploy)
- Object storage + CDN setup

### Phase 2 — Data Tier (Week 4)

- MySQL primary (managed service preferred)
- 2 read replicas
- Backup automation + PITR
- Redis cluster

### Phase 3 — Application Tier (Weeks 5-6)

- Auto-scaling group with launch template
- Load balancer + health checks
- Migrate DB schema + seed catalog
- Frontend build → CDN

## Phase 4 — Integrations (Week 7)

- GSB adapter deploy + IP whitelist
- Nashmi webhook validation
- Payment gateway integration + testing
- SMS + Email providers

## Phase 5 — Observability (Week 8)

- Log aggregation
- Metrics + dashboards
- Alerting rules + on-call rotation
- Runbook for common failures

## Phase 6 — Hardening (Week 9)

- WAF rule tuning (start permissive, tighten)
- Penetration testing
- Load testing (target 500 concurrent users)
- DR drill

## Phase 7 — Go-Live (Week 10)

- Soft launch (whitelist users)
- Monitor tightly for 2 weeks
- Full public launch

# 11. Key Decisions to Confirm

Before finalizing, please decide:

| Decision       | Options                                                        |
|----------------|----------------------------------------------------------------|
| Cloud provider | AWS / Azure / GCP / on-premises (Hostinger unlikely to scale)  |
| Data region    | Jordan (data-sovereignty) or Middle East region (Bahrain, UAE) |

| Decision           | Options                                                        |
|--------------------|----------------------------------------------------------------|
| DB managed vs self | RDS/Aurora (recommended) vs self-managed MySQL                 |
| Object storage     | AWS S3 vs Backblaze vs self-hosted MinIO                       |
| Payment gateway    | eFAWATEERcom / Arab Bank / Cliq — depends on NGO agreement     |
| SMS/Email provider | Local (Zain/Umniah) vs global (Twilio/SES)                     |
| Monitoring stack   | CloudWatch (managed) vs Prometheus+Grafana (self-hosted)       |
| DR strategy        | Warm standby (recommended for RTO=1h) vs cold backup (RTO=8h+) |

## 12. Comparison: SPAC Reference Architecture vs This Proposal

| Aspect        | SPAC (Reference)               | This Proposal                        | Why Different                            |
|---------------|--------------------------------|--------------------------------------|------------------------------------------|
| App servers   | 2 fixed                        | 3-10 auto-scale                      | JEA scale requires horizontal elasticity |
| Database      | Single MySQL Primary           | Primary + 2 read replicas + PITR     | 90k users need read scaling + HA         |
| File storage  | Not shown (implied filesystem) | Object storage + CDN                 | 1.2M docs/y kills local filesystem       |
| Cache/Session | Not shown                      | Redis cluster                        | Sessions + rate limits + queue           |
| Queue         | Not shown                      | Horizon workers                      | Cron jobs + async notifications          |
| CDN           | Not shown                      | Cloudflare/CloudFront                | Public cert verification traffic         |
| Observability | Not shown                      | ELK + Prometheus + Grafana           | Production needs visibility              |
| Integrations  | SMS + External SQL + Job API   | GSB + Nashmi + Payment + SMS + Email | JEA's real dependencies                  |
| DR            | Not shown                      | Cross-region async + failover        | Regulatory + business continuity         |

| Aspect | SPAC (Reference) | This Proposal       | Why Different         |
|--------|------------------|---------------------|-----------------------|
| CI/CD  | Not shown        | Blue/Green pipeline | Zero-downtime deploys |

### 13. Handoff Checklist for Implementation

- ☐ Cloud account provisioned + billing alerts
- ☐ Domain + DNS configured
- ☐ SSL certificates issued
- ☐ VPC + network isolation established
- ☐ Database provisioned + backups tested
- ☐ Redis cluster verified
- ☐ Object storage bucket created + IAM configured
- ☐ Load balancer + health checks operational
- ☐ Application auto-scale group + launch template
- ☐ CI/CD pipeline (GitHub → deploy)
- ☐ External integrations tested end-to-end
- ☐ Observability stack ingesting from all layers
- ☐ Alerting routed to on-call
- ☐ Runbook written for top 5 failure modes
- ☐ Load test completed at 2× expected peak
- ☐ DR drill executed successfully
- ☐ Security review + pen test complete
- ☐ Documentation handed to operations team

**End of document.**  
Prepared by ESP v2 architecture team, 2026-07-25.  
Sources: docs/architecture/\* , docs/manual-summary.md , docs/EDA\_DECISION\_CHAIN.md , SPAC reference diagram.